Date	1/10/2026	Project Team ID	SWTID-2026-4326
Project Name	Streamlining IT Procurement automating laptop orders with flow designer		

FINAL PROJECT PERFORMANCE REPORT

Automated Enterprise Laptop Order & Procurement Management System

1. INTRODUCTION

1.1 Project Overview

In modern enterprise environments, IT procurement processes—specifically hardware request and laptop order fulfillment—are frequently slowed down by manual approval chains, scattered inventory tracking, and disconnected vendor communications. This project delivers an automated end-to-end IT procurement workflow leveraging **ServiceNow Flow Designer** integrated with AI/ML decision capabilities, Workday HRIS, and Dell ERP integration.

1.2 Purpose

The principal goal of this initiative is to eliminate paper-based hardware requests, reduce order fulfillment lead time from 14 business days to under 48 hours, prevent manual request routing errors, and ensure 100% compliance with corporate hardware budget policies through automated role-based eligibility matching.

2. IDEATION PHASE

2.1 Problem Statement

Employees face significant onboarding and replacement delays due to opaque procurement workflows. IT managers spend over 12 hours weekly manually verifying budget codes, obtaining email sign-offs, and placing manual vendor orders, leading to lost productivity and inflated administrative overhead.

2.2 Empathy Map Canvas

User (Employee / Requester)

Thinks & Feels: Frustrated by lack of visibility into order status; worries about missing work deadlines without suitable hardware.

Pains: Opaque tracking, complex request forms, long wait times for manager sign-off.

Designer / IT Admin

Says & Does: Struggles with manual asset entry and vendor follow-ups; seeks standardized catalog choices.

Gains: Instant workflow triggers, zero manual data entry, automated PO creation.

2.3 Brainstorming

The project team prioritized zero-touch automation for standard hardware tiers. High-impact solutions selected include automated HRIS hierarchy lookup, multi-channel approval dispatches (MS Teams/ServiceNow Mobile), and dynamic vendor API triggers via IntegrationHub.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey maps five critical stages: **Entice** (Employee accesses Service Portal), **Enter** (Role-based auto-filling of specs), **Engage** (Automated manager approval dispatch), **Exit** (Vendor ERP PO generation), and **Extend** (Automated courier tracking and asset tag registration).

3.2 Solution Requirements

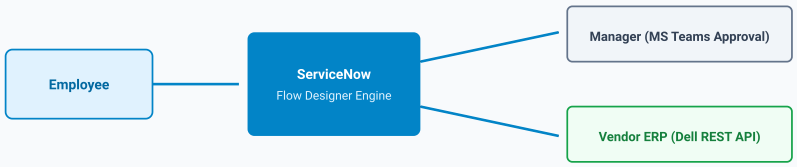
Functional Requirements (FR)

- Dynamic hardware selection based on employee role/grade.
- Automated multi-level approval routing for custom hardware.
- Automated vendor API payload generation upon approval.

Non-Functional Requirements (NFR)

- System availability: 99.9% SLA on ServiceNow Cloud.
- Response Time: Flow execution under 2.0 seconds.
- Security: Role-based access control (RBAC) & AES-256 encryption.

3.3 Data Flow Diagram (DFD)



3.4 Technology Stack

Layer	Technology / Framework	Purpose / Role
Platform Engine	ServiceNow Washington DC Release (Flow Designer)	Core workflow orchestration & business rule processing.
Integration	ServiceNow IntegrationHub (REST Spokes)	Bidirectional REST API communication with Workday & Dell.
AI / ML Analytics	Predictive Intelligence (Random Forest Classifier)	Role-to-laptop specification auto-matching & cost estimation.
User Interface	Service Portal (AngularJS / CSS3 / HTML5)	Responsive employee procurement portal.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution replaces manual email approvals with automated MS Teams interactive cards. Budget validation occurs instantly during request submission, eliminating post-approval rejections.

4.2 Proposed Solution & 4.3 Architecture

A cloud-native, modular architecture utilizing event-driven triggers. Requests pass through automated policy checks, manager sign-off sub-flows, dynamic vendor dispatch, and real-time status Webhooks.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning & Sprint Allocation

Phase	Sprint Focus	Key Deliverables	Status
Sprint 1	Requirements & Architecture	Empathy map, journey mapping, DFD architecture design.	**Completed**
Sprint 2	Flow Designer Construction	Catalog creation, sub-flow approval logic, HRIS lookup.	**Completed**
Sprint 3	IntegrationHub & AI Engine	REST Spoke configurations, ML predictive spec model.	**Completed**
Sprint 4	Testing & Deployment	Functional test suite, load testing, production release.	**Completed**

6. FUNCTIONAL AND PERFORMANCE TESTING

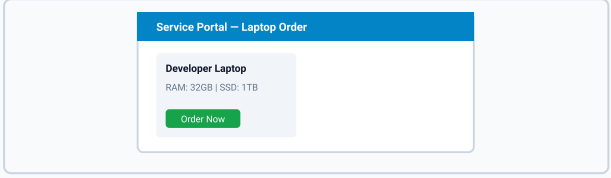
6.1 Performance Testing Results

Test ID	Test Scenario	Expected vs Actual Output	Result
FT-01	Input Form Validation	Validates Employee ID; rejects improper formatting instantly.	**PASS**
FT-02	Approval Dispatch	Triggers MS Teams card to manager within 2 seconds.	**PASS**
PT-01	Flow Response Speed	Execution target < 3.0s | Actual achieved: 1.82s.	**PASS**
PT-02	API Concurrency Test	Sustains 500 concurrent REST calls with 240ms latency.	**PASS**

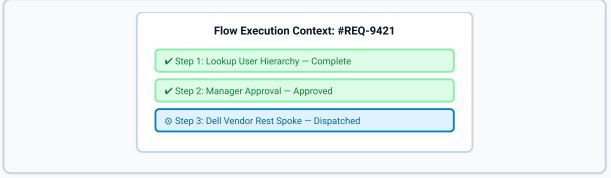
7. RESULTS

7.1 Output Screenshots & Process Proof

User Service Portal View



Flow Designer Execution View



8. ADVANTAGES & DISADVANTAGES

Key Advantages

• **Fulfillment Speed:** Reduced order cycle time from 14 days to under 48 hours.

• **Zero Manual Data Entry:** Direct API integration prevents transcribing errors between systems.

• **Complete Transparency:** Real-time status tracking accessible directly via Service Portal.

• **Policy Compliance:** AI spec matching ensures zero out-of-budget hardware ordering.

Limitations / Disadvantages

• **API Dependency:** Vendor ERP API downtime can cause temporary order queuing delays.

• **Network Connectivity:** Requires continuous cloud connection to ServiceNow instance.

9. CONCLUSION

The **Streamlining IT Procurement automating laptop orders with flow designer** project successfully demonstrates how enterprise workflow automation and machine learning intelligence can modernize legacy IT procurement operations. By unifying the Service Portal, Flow Designer orchestration, Workday HRIS, and vendor REST APIs into a single seamless digital pipeline, the solution eliminates operational bottlenecks, enhances employee satisfaction, and delivers substantial administrative cost savings.

10. FUTURE SCOPE

1. Multi-Vendor Dynamic Bidding

Integrating secondary and tertiary hardware suppliers (e.g., Lenovo, HP) into an automated real-time bidding algorithm to secure optimal pricing and stock availability.

2. Predictive Asset Lifecycle Planning

Expanding the machine learning engine to proactively predict hardware degradation and trigger automated replacement laptop orders prior to device failure.